HDE Note

1 Basic Knowledge & SVCDE Methods for ODE

1.1 Linear Algebra || Calculus || SVCDE Method for ODE

Matrix Calculate: $det(kA) = k^n det(A)$ || $det(A^{-1}) = det(A)^{-1}$ || $(AB)^* = A^*B^*$ || $det(A) = det(A^T)$.

· **Eigenvalue λ & Eigenvector $\mathbf{x}$**: Def: $A\mathbf{x} = \lambda\mathbf{x}$. || Method: $^1 det(A-\lambda I) = 0 \to \lambda$. $^2 (A-\lambda I)\mathbf{x} = 0 \to \mathbf{x}$.

· **Arithmetic Manipulate (AM) λ**: λ 重复出现的次数. $\geq$ **Geometric Manipulate (GM) of λ**: # $\mathbf{x}$ under λ.

· **Generalised Eigenvectors**: If AM>GM: Using $\xi_\lambda = (A-\lambda\mathbb{I})\eta_1, \eta_{n-1} = (A-\lambda\mathbb{I})\eta_n$ for each ξ_λ. η_j are *generalised eigenvectors*.

Some Defs: $\nabla F = (\partial_x F, \partial_y F, \partial_z F, ...)$ || **Simply Connected**: 没有洞, 相互连接的区域. || ∂D: 区域 D 的边界.

Method of Integrating Factors: $y' + p(x)y = q(x)$. $\Rightarrow$ Integrating Factor:$\mu = e^{\int p(x)dx}$, $\mu\, y = \int q(x)\cdot\mu\, dx$

Homogeneous 2nd ODE form: $ay'' + by' + cy = 0$ $\Rightarrow$ Characteristic $ar^2 + br + c = 0$, Roots: r_1, r_2 ps:$y'' \pm \lambda y = 0$ 见附录

· $^1\Delta > 0, y = C_1e^{r_1x} + C_2e^{r_2x}$; $^2\Delta = 0, y = (C_1 + C_2x)e^{rx}$; $^3\Delta < 0, y = e^{\alpha x}[C_1cos(\beta x) + C_2sin(\beta x)]$ ps: $r_1, r_2 = \alpha \pm \beta i$

non-Homogeneous 2nd ODE form: $ay'' + by' + cy = g(x)$ $\Rightarrow$ $y = y_h + y_p$ ps:y_h is sol for $g(x) = 0$.

· Method of Undetermined Coefficients: Let $A_n(x) = \sum_{i=0}^n a_ix^i$, $B_n(x) = \sum_{i=0}^n b_ix^i$, $P_n(x) = \sum_{i=0}^n p_ix^i$.

1. If $g(x) = P_n(x) \Rightarrow y_p = x^sA_n(x)$, $s :=$ $0, (0 \neq r_1, r_2)$; $1, (0 = r_1\ or\ r_2)$; $2, (0 = r_1 = r_2)$.
2. If $g(x) = Me^{kx} \Rightarrow y_p = Cx^se^{kx}$, $s :=$ $0, (k \neq r_1, r_2)$; $1, (k = r_1\ or\ r_2)$; $2, (k = r_1 = r_2)$.
3. If $g(x) = Mcos(kx) + Nsin(kx) \Rightarrow y_p = x^s(C_1cos(kx) + C_2sin(kx))$, $s :=$ $0, (r_1, r_2 \neq ik)$; $1, (r_1, r_2 = ik)$.
4. If $g(x)$ = product from 1,2,3. $\Rightarrow$ y_p = corresponding product of y_p (x^s 只乘一次) , $s :=$ 令 y_p 与 y_c 没有相同项. (≤ 2)

Euler Equation: General: $a_nx^ny^{(n)} + a_{n-1}x^{n-1}y^{(n-1)} + \cdots + a_0y = 0$ $\Rightarrow$ Seek Solution as: $y = x^\alpha$ $\Rightarrow$ solution: $y = \sum_i c_ix^{\alpha_i}$

1.2 Linear || (ODE) Uniqueness Theorem || Autonomous

Def of Linear: Let L 去掉 $y_i^{(n)}$ 的函数 (e.g. $L = \frac{d^2}{dx^2} - x$ for $y'' - xy = x^2$)

$\Rightarrow$ Linear if: $L[a_1y_1 + a_2y_2] = a_1L[y_1] + a_2L[y_2]$

Existence & Uniqueness Theorem (Real func):

· $^1\mathbf{y}'(t) = \mathbf{F}(t, \mathbf{y}^T), \mathbf{y}(t_0) = \mathbf{y_0}$. Let $R_i = [t_0 - \delta, t_0 + \delta] \times \prod_j [y_{0_j} - \delta_j, y_{0_j} + \delta_j]$.

· $^2 F_j$, $\partial_{y_i}F_j : R_i \to \mathbb{R}$ are continuous. $\Rightarrow$ $\exists$ unique solution $\mathbf{y}$ in R_i'

$R_i' = [t_0 - \varepsilon, t_0 + \varepsilon] \times \prod_j [y_{0_j} - \varepsilon_j, y_{0_j} + \varepsilon_j] \subseteq R_i$.

Autonomous: For $\mathbf{y}'(t) = \mathbf{f}(t, \mathbf{y}^T)$. $\partial_x\mathbf{f} = \mathbf{0} \Rightarrow$ autonomous. or: $\mathbf{f}(t, \mathbf{y}^T) = \mathbf{f}(\mathbf{y}^T)$

2 Linear ODEs

n-th ODE → 1st ODEs: For $y^{(n)} = F(t, y, y', ..., y^{(n-1)})$. $\Rightarrow$ Let $x_i = y^{(i-1)}$.

Def of Linear 1st ODEs: $\mathbf{y}'(t) = P(t)\mathbf{y}(t) + \mathbf{g}(t)$ $(\star)$, P is matrix-valued function.

Homogeneous: $\mathbf{g}(t) = \mathbf{0}$.

Existence & Uniqueness Theorem: If P_{ij}, $\mathbf{g}$ of $(\star)$ continuous in R, $\mathbf{y}(t_0) = \mathbf{y}_0$

$\Rightarrow$ $\exists$ unique solution $\mathbf{y}$ in R' $R' = [t_0 - \varepsilon, t_0 + \varepsilon] \subseteq R = [t_0 - \delta, t_0 + \delta]$

2.1 Homogeneous

Principle of Superposition: If $\mathbf{y}_i$ are solutions $\Rightarrow$ $\sum_i c_i\mathbf{y}_i$ also. || $S := \{\mathbf{y}|\mathbf{y}' = P\mathbf{y}\}$ is a *linear space*, $dim(S) = n$ if P_{ij} continuous.

Fundamental Set of Solutions (FSoS): If $\{\mathbf{y}^{(j)}(t)\}$ *linearly independent* at $t \in I$, it is FSoS in I.

General Solution: If $\{\mathbf{y}^{(j)}(t)\}_{j=1}^n$ is FSoS (Linearly Independent) $\Rightarrow$ General solution: $\mathbf{y} = \sum_{j=1}^n c_j\mathbf{y}_j$

2.1.1 Eigenvalue/vector Method (A constant)

AM=GM|Real λ: general solution: $\mathbf{y} = \sum_j c_je^{\lambda_jt}\xi_j$.

AM=GM|Complex λ: (ctd.) For $\lambda, \bar{\lambda} \in \mathbb{C}$. $\Rightarrow$ $e^{\lambda t}\xi = \mathbf{u} + \mathbf{v}i$. $\Rightarrow$ Independent sol: $\mathbf{u}, \mathbf{v}$ $\Rightarrow$ General solution: $\mathbf{y} = c_1\mathbf{u} + c_2\mathbf{v} + \cdots$

AM>GM: (ctd.) (For each ξ_λ, under λ) $\mathbf{y}^{(1)} = e^{\lambda t}\xi_\lambda$ $\mathbf{y}^{(2)} = e^{\lambda t}(t\xi_\lambda + \eta_1)$ $\mathbf{y}^{(3)} = e^{\lambda t}(\frac{t^2}{2!}\xi_\lambda + t\eta_2 + \eta_1)$, ... 类似泰勒展开

2.1.2 Matrix Method | Exponential

Matrix Method: $\mathbf{y}(t) = e^{At}\mathbf{y}(0)$ (分两种方法: 1. 直接展开; 2. 通过一些已有 property 计算. ps: $e^{At} = \Psi$)

· **Property|AM=GM**: $e^{At} = Te^{Dt}T^{-1}$, $T = [\xi_1, ..., \xi_n]$, $D = diag(\lambda_1, ..., \lambda_n)$, $e^{Dt} = diag(e^{\lambda_1t}, ..., e^{\lambda_nt})$

· **Property|AM>GM**: $e^{At} = Te^{Jt}T^{-1}$, $T = [\xi_1, ..., \xi_i, \eta_1^{(i)}, ..., \xi_j, \eta_1^{(j)}, ...]$, $J = T^{-1}AT$ (Jordan form) , e^{J_it} = 见右边

Diagonalisation Method: (Ctd.) Let $\mathbf{y} = T\mathbf{x}$ $\Rightarrow$ $\mathbf{x}' = J\mathbf{x}$ [or $\mathbf{x}' = D\mathbf{x}$] 按前面方法解

2.1.3 Matrix Method | Fundamental Matrix

Fundamental matrix Ψ: If $\{\mathbf{y}^{(j)}(t)\}$ is FSoS. $\Rightarrow$ $\Psi(t) = [\mathbf{y}^{(1)}(t), ..., \mathbf{y}^{(n)}(t)]$.

· **Property**: $\Psi'(t) = P(t)\Psi(t)$ (It's a sol). || $det[\Psi(t)] = W(t) \neq 0$ || general solution: $\mathbf{y} = \Psi\mathbf{c}$, $\mathbf{c} = \Psi^{-1}(t_0)\mathbf{y}_0$

· **Relation with e^{At}**: $e^{At} = \Psi(t)$ with $\Psi(0) = \mathbb{I}$ || $e^{At} = \Psi(t)\Psi^{-1}(0)$

Solve: $\mathbf{y}' = P\mathbf{y}$ || If P constant, $P = A$ || λ, ξ, η be 特征值/向量/广义特征向量.

Abel's Theorem: If $W(t_0) \neq 0, t_0 \in I$ $\Rightarrow$ $\forall t \in I, W(t) \neq 0$

Abel's Formula: $W(t) = W(t_0)e^{\int_{t_0}^t tr[P(s)]ds}$; $t_0, t \in I$ $P(t)$ is continuous in I.

Wronskian at x: $W[\mathbf{y}^{(1)}, ..., \mathbf{y}^{(n)}](t) = \begin{vmatrix} y_1^{(1)}(t) & \cdots & y_1^{(n)}(t) \\ \cdots & \cdots & \cdots \\ y_n^{(1)}(t) & \cdots & y_n^{(n)}(t) \end{vmatrix}$.

· $W(t) \neq 0, \forall t$ $\Rightarrow$ $\{\mathbf{y}^{(j)}\}$ **linearly independent**. ps: 经常用 Abel Theorem

$$e^{J_it} = e^{\lambda_it}\begin{pmatrix} 1 & t & \frac{t^2}{2} & \cdots & \frac{t^{n-1}}{(n-1)!} \\ 0 & 1 & t & \cdots & \frac{t^{n-2}}{(n-2)!} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix} \quad \text{ps: (e.g.) } J_3 = \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix}$$

Exponential of Matrix: $e^{At} := \sum_{n=0}^\infty \frac{(At)^n}{n!} = \mathbb{I} + At + \frac{1}{2!}A^2t^2 + \cdots$

$:= \lim_{n\to\infty}(\mathbb{I} + \frac{At}{n})^n$

· **Property**: $e^{A+B} = e^Ae^B$ if $AB = BA$ || $e^{A\cdot 0} = \mathbb{I}$ || $[e^{At}]' = Ae^{At}$

2.2 non-Homogeneous

Theorem: Solution of $(\star)$: $\mathbf{y} = \sum_i c_i\mathbf{y}^{(i)} + \mathbf{y}_p$, where $\mathbf{y}^{(i)}$ is solution of it's Homogeneous. ⇓通常会让特解中积分后的常数 c=0

Variation of Parameters: $\mathbf{y} = \mathbf{y}_g + \mathbf{y}_p$; $\mathbf{y}_g = \Psi(t)\Psi^{-1}(t_0)\mathbf{y}(t_0)$; $\mathbf{y}_p = \Psi(t)\mathbf{u}(t), \mathbf{u}(t) = \int_{t_0}^t \Psi^{-1}(s)\mathbf{g}(s)ds$. 经常 $\Psi(t)\mathbf{u}' = \mathbf{g}$ 来找 $\mathbf{u}$

Matrix methods|Diagonalisation: ^{1}Let $\mathbf{y} = T\mathbf{x}$ $^2(\star) \Rightarrow \mathbf{x}' = D\mathbf{x} + T^{-1}\mathbf{g} = D\mathbf{x} + \mathbf{h}$ ^{3} 直接计算; $x_i = c_ie^{r_it} + \underbrace{e^{r_it}\int_{t_0}^t e^{-r_is}h_i(s)ds}_{y_p}$

Undetermined Coefficients: when $\mathbf{g}(t)$ is polynomials, or real/complex exponential. Let $P_n(t) = \sum_{i=0}^n \mathbf{p}_it^i, A_n(t) = \sum_{i=0}^n \mathbf{a}_it^i$

^{1} 构造 $\mathbf{y_p}$ 与 $\mathbf{g}$ 结构相匹配.(e.g. If $\mathbf{g} = P_n(t) \Rightarrow \mathbf{y}_p = A_n(t)$; If $\mathbf{g} = \mathbf{c}e^{kt} \Rightarrow \mathbf{y}_p = \mathbf{a}e^{kt}$) ^{2} 特殊: If $\mathbf{g} = \mathbf{c}e^{\lambda t}$,$\lambda$ is eigenvalue with $AM = n \Rightarrow \mathbf{y}_p = e^{\lambda t}\sum_{i=0}^n \mathbf{a}_it^i$ ^{3} 带入原方程得未知系数

Inversse of Matrix: $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$

$(e.g.) J_3^{-1} = \frac{1}{\lambda}\begin{pmatrix} 1 & -\frac{1}{\lambda} & \frac{1}{\lambda^2} \\ 0 & 1 & -\frac{1}{\lambda} \\ 0 & 0 & 1 \end{pmatrix}$

2.3 Geometrical Interpretation & Linear Approximation

* 此节中，Consider nonlinear autonomous system: $x' = F(x, y), y' = G(x, y)$ $(**)$

Stable: Critical Point $\mathbf{x}_0$ is stable iff $\forall\varepsilon > 0, \exists\delta > 0$ s.t. $\forall$ sols $\mathbf{x}(t)$ with $||\mathbf{x}(0) - \mathbf{x}_0|| < \delta$, then $||\mathbf{x}(t) - \mathbf{x}_0|| < \varepsilon, \forall t \geq 0$ || 直观理解: 开始近 → 保持近.

Attracting: Critical point $\mathbf{x}_0$ is attracting iff $\exists\delta > 0$ s.t. $\forall$ sols $\mathbf{x}(t)$ with $||\mathbf{x}(0) - \mathbf{x}_0|| < \delta$ satisfies: $\lim_{t\to\infty}\mathbf{x}(t) = \mathbf{x}_0$ || 直观理解: 开始近 → 不断趋向.

Asymptotically Stable: Critical point $\mathbf{x}_0$ is asy.stab iff ^{1} Stable, ^{2} Attracting.

Theorem: If $\mathbf{x}_*(t) = (x_*, y_*)$ s.t. $F(\mathbf{x}_*), G(\mathbf{x}_*) \neq 0$. $\Rightarrow$ $\exists H$ s.t. $H(x, y) = (\tilde{x}, \tilde{y}), \tilde{x}' = 1, \tilde{y}' = 0$ H is a smooth change of variable function in neighbourhood of $\mathbf{x}_*$

Non-Linear → Linear: For system $(**)$, assume $\mathbf{x}_0 = (x_0, y_0)$ is critical point. $\Rightarrow$ $\mathbf{u} = \mathbf{x} - \mathbf{x}_0$, then $\mathbf{u}' \approx A\mathbf{u}$ Linear; $A = \begin{pmatrix} \partial_xF(\mathbf{x}_0) & \partial_yF(\mathbf{x}_0) \\ \partial_xG(\mathbf{x}_0) & \partial_yG(\mathbf{x}_0) \end{pmatrix}$ Jacobian

Phase Plane: O_{xy} referred as phase plane.

Trajectories: Sol of ODE. $\mathbf{x}(t) = (x(t), y(t))^T$

Phases Portrait: 一组有代表性的轨迹

Critical Point (Equilibrium Point):

$[x' = F(\mathbf{x}_0) = 0, y' = G(\mathbf{x}_0) = 0]$ or $[\mathbf{x}' = \mathbf{0}]$

(critical point 外变化简单)

2.4 Classification of Critical Points

* 此节中，Consider linear system: $\mathbf{x}' = A\mathbf{x}$. Critical Point(s): when $\mathbf{x}' = A\mathbf{x} = \mathbf{0}$

Classification of Critical Points at (x_0, y_0) (Linear): r_1, r_2 be sol of $det(A - \lambda I) = 0$. || $\mathbb{C}: r = \lambda \pm i\mu(\mu > 0)$ || If A constant, write sol: $\mathbf{x} = c_1e^{r_1t}\xi_1 + c_2e^{r_2t}\xi_2$ || $GM = 1$: $\mathbf{x} = c_1e^{rt}\xi + c_2e^{rt}(t\xi + \eta)$

$\mathbb{R}/\mathbb{C}$	Condition || Stability	Type || Name	Phase Plane Description	Other	
$\mathbb{R}$	$r_1 < r_2 < 0$ || asy.stab	N || NSk	向原点,ξ_2 直线,ξ_1 曲线,and ξ_1 周围 $y = \pm x^3$	$c_2 \neq 0, t \to \infty$:ξ_2 主导方向; $c_2 = 0, t \to \infty$:ξ_1 主导方向	PS: **N** = Node **PN** = Proper Node **IN** = Improper or: Degenerate Node **SP** = Saddle Point **SpP** = spiral point or: Focus Point **C** = Center **NSk** = Nodal Sink **NSo** = Nodal Source
	$r_1 > r_2 > 0$ || unstable	N || NSo	原点向外,ξ_2 直线,ξ_1 曲线,and ξ_1 周围 $y = \pm x^3$	$c_1 \neq 0, t \to \infty$:ξ_1 主导方向; $c_1 = 0, t \to \infty$:ξ_2 主导方向	
	$r_1 > 0 > r_2$ || unstable	SP || SP	$t \to \infty$,ξ_1 从原点向外,ξ_2 从外向原点 and: 像 $y = \pm\frac{1}{x}$, 同进同出	$t \to \pm\infty : \|\mathbf{x}\| \to \infty$; $t \to \infty : c_1, c_2 \neq 0, \|\mathbf{x}\| \to \infty$,$\xi_1$ 主导; $t \to \infty : c_2 = 0, \|\mathbf{x}\| \to \infty$,$\xi_1$ 主导; $t \to \infty : c_1 = 0, \|\mathbf{x}\| \to 0$,$\xi_2$ 主导	
	$r_1 = r_2 < 0$, GM=2 || asy.stab	PN || PN or Stable Star	直线 向原点	直线, u_1/u_2 is t independent	
	$r_1 = r_2 > 0$, GM=2 || unstable	PN || PN or Unstable Star	直线 从原点向外	直线, u_1/u_2 is t independent	
	$r_1 = r_2 < 0$, GM=1 || asy.stab	IN (AL:Type: SpP) || IN (Stable)	S 曲线, 向原点	$t \to \infty, \|\mathbf{x}\| \to 0, \xi$ 主导 ps: 旋转方向大体和 $\eta + c_2\xi$ 方向相同	
	$r_1 = r_2 > 0$, GM=1 || unstable	IN (AL:Type: SpP) || IN (Unstable)	S 曲线, 从原点向外	$t \to \infty, \|\mathbf{x}\| \to \infty, \xi$ 主导 ps: 旋转方向大体和 $\eta + c_2\xi$ 方向相同	
$\mathbb{C}$	$\lambda \neq 0, \lambda > 0$ || unstable	SpP || Unstable Focus	向外椭圆 (elliptical) 螺旋	$t \to \infty, \|\mathbf{x}\| \to \infty$ ps: 考虑 $A = (a, b; c, d)$, 如果 bc>0, 顺时针，如果 bc<0, 逆时针	
	$\lambda \neq 0, \lambda < 0$ || asy.stab	SpP || Stable Focus	向内椭圆 (elliptical) 螺旋	$t \to \infty, \|\mathbf{x}\| \to 0$ ps: 考虑 $A = (a, b; c, d)$, 如果 bc>0, 顺时针，如果 bc<0, 逆时针	
	$\lambda = 0$ || stable (AL:Indeterminate)	C (AL:C or SpP) || C	椭圆 (elliptical) and 半长轴 ξ 实部方向	Bounded trajectory or $\exists$ Periodic Trajectories	

AL: Almost Linear Sys. || For Almost Linear systems: $\mathbf{u}' = A\mathbf{u} + \underline{\eta}$, 如果 linear Approximation (去掉 $\underline{\eta}$), 按照 Linear 考虑. 如果不去 $\underline{\eta}$, 按照去掉的算 r_1, r_2, Type 按照正常 / (AL) 的来.(图像会到 $(0, 0)$)

Some Defs: $Separatrix$: 将不同区域分开的轨迹; $Basin\ of\ attraction$: 所有趋近 critical point 的轨迹. ps: In any nonlinear ODEs, it's important to determine the *basin of attraction*.

3 Non-Linear ODE Methods

* 此节中，Consider two-dimensional autonomous system: $x' = F(x, y), y' = G(x, y)$ and $\mathbf{x} = (x, y)^T$ (#) Lyapunov's: 不需要知道微分方程的确切解 critical value. Poincaré-Bendixson: 寻找 limit cycle

Lyapunov Stability Theorem: ^{1}(#); ^{2}ICP $\mathbf{x}_0 \in D$; $^3\exists$ 正定且连续 $E : D \subset \mathbb{R}^2 \to \mathbb{R}, E(\mathbf{x}_0) = 0$; || $^4 \frac{dE}{dt} = \nabla E\cdot\mathbf{x}' = \partial_xE\cdot F + \partial_yE\cdot G$ 负 (半) 定 on $D \setminus \mathbf{x}_0(D) \Rightarrow \mathbf{x_0}$ asp.(stab)

· **Lyapunov function**: 符合上面 1-4 条件的

Lyapunov Instability Theorem: ^{1}(#); ^{2}ICP $\mathbf{x}_0 \in D$; $^3\exists E : D \subset \mathbb{R}^2 \to \mathbb{R}, \partial_iE$ 连续,$E(\mathbf{x}_0) = 0$; $^4\forall$ 领域$\mathbf{x}_0, \exists\mathbf{x}_1$ $E(\mathbf{x}_1) > 0$; || $^5 \frac{dE}{dt}$ 正定 on $D \Rightarrow \mathbf{x_0}$ unstable

Periodic Solution: Soln $\mathbf{x}$ has: $\mathbf{x}(t + T) = \mathbf{x}(t), \forall x$. || 闭合曲线 ⇔ Periodic sol || Periodic Solution + 附近有吸引或排斥 ⇒ Limit Cycle || when $\dot{r} = 0, \dot{\theta} = const$. Periodic sol

Limit cycle: Limit cycle ⇒ Periodic Solution || Def $t \to \pm\infty$, $\exists$ non-closed trajectory asymptotes to it. (inside or outside) ps: It's Orbital Stability

1. Limit cycle is: ^{1}asy.stab: $\forall$ trajectories asy. it; ^{2}semistable: $\exists$ trajectories 单方向进/出; ^{3}unstable: $\forall$ trajectories away from it.
2. **Theorem (If $\exists$ limit cycle / closed)**: $^1F, G$ 一阶偏导连续 in simply connected D; $^2\exists$ 极限环/封闭环 $(\star)$ in D. $\Rightarrow$ $^1\exists$ CP in $(\star)$; ^{2}If $\exists$! CP in $(\star)$, it's not SP.
3. **Theorem (No limit cycle / closed)**: $^1F, G$ 一阶偏导连续 in simply connected D; $^2\partial_xF + \partial_yG$ 符号不变 on D $\Rightarrow$ No closed trajectory in D.

   2 的推论: ^{1}If simple connected D no CP $\Rightarrow$ No closed trajectory in D. ^{2}If simple connected D has unique CP (Type:SP) $\Rightarrow$ No closed trajectory in D

Poincaré-Bendixson Theorem: $^1F, G$ 一阶偏导连续 in D; $^2\exists D_1 \subset D$ bounded; $^3R = D_1 \cup \partial D_1$ has NO CP; || ^{4} $\exists$ soln $\mathbf{x} \in R, \forall t \geq t_0$ $\Rightarrow$

$\Rightarrow$ $^1\exists$ closed trajectory. ^{2} Either $\mathbf{x}$ *sprial towards a Periodic trajectory.* or $\mathbf{x}$ *periodic* ^3R is not simply connected

Find Trapping Region / Limit Cycle: **trapping region**: 符合上面定理 2-4 除 CP 的 region. Method: *polar coordinate*, 计算 $r^2 = x^2 + y^2$ 对 t 的求导, 寻找 $r\dot{r} \geq 0$ or $r\dot{r} \leq 0$

Positive (Negative) [semi-]definite:

$E : D \subset \mathbb{R}^2 \to \mathbb{R}$ is positive (negative) [semi-]definite iff:

$^1E(\mathbf{x}_0) = 0$

$^2\forall\mathbf{x} \in D, \mathbf{x} \neq \mathbf{x}_0, E(x, y) > 0$ or $(E(x, y) < 0)$ or $[E(x, y) \geq 0]$

ps: Neighbourhood 领域

Isolated Critical Point: Critical Point $\mathbf{x}_0$ *isolated* iff:

$\exists$ Neighbourhood containing no other critical points.

4 Fourier Series

If $y'' - \lambda y = 0, y(0), y(L) = 0$. The Solution of this SL Problem will be a Fourier Series

* BVPs 可能没有解/一个解/无数解; IVP 只会有一个解

4.1 Function Space, Eigenfunctions, Periodic Function and Orthogonal Function

Periodic Functions: Function $f(x)$ is Periodic iff $f(x + T) = f(x), \forall x$. **Period**: T. **Fundamental Period**: The smallest possible T. || **Property**: If f, g has same period T. $\Rightarrow$ $af(x) + bg(x)$ also has period T.

Differential Operators: Any linear combination of n-th derivatives defines a linear map or linear operator. e.g. $p(x)\frac{d}{dx} + \frac{d^2}{dx^2} : y \mapsto p(x)\frac{dy}{dx} + \frac{d^2y}{dx^2}$

Eigenvalues & Eigenfunctions: If D is a *differential operator* $\Rightarrow$ If $Df = \lambda f$, then λ is **Eigenvalue** and f is **Eigenfunctions** || **Property**: 对于微分方程: $Df = \lambda f$ $\Rightarrow$ solution: $f = \sum c_nf_n$ f_n are Eigenfunctions

Inner Product: $(f, g) = \int_a^b f(x)\overline{g(x)}dx$ For funcs f, g in (a, b) || **Orthogonal**: $(f, g) = 0$ || **Square Norm**: (f, f). || **Kronecker delta**: $\delta_{mn} = \{0, m \neq n;\ 1, m = n$.

4.2 Fourier Series

Fourier Series: Periodic function with $T = 2L$, $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left(a_n cos(\frac{n\pi x}{L}) + b_n sin(\frac{n\pi x}{L})\right)$ (*)

· **Euler-Fourier Formulas**: Suppose (*) converges in the interval $[-L, L]$. Then: $a_n = \frac{1}{L}\int_{-L}^{L} f(x)cos(\frac{n\pi x}{L})dx$ $b_n = \frac{1}{L}\int_{-L}^{L} f(x)sin(\frac{n\pi x}{L})dx$ $a_0 = \frac{1}{L}\int_{-L}^{L} f(x)dx$ (Average Value of f over $2L$)

Piecewise Continuous: $f(x)$ piecewise continuous on $[a, b]$ iff $\exists a = x_0 < \cdots < x_n = b$ s.t. [1] $f(x)$ continuous $\forall x \in (x_{i-1}, x_i)$ [2] $f(x_i^{\pm}) = \lim_{x \to x_i^{\pm}} f(x)$ finite.

Fourier Convergence Theorem: If [1] f, f' piecewise continuous on $[-L, L]$, [2] $f(x + 2L) = f(x)$. $\Rightarrow$ Fourier Series of f converges. **Fourier 和 SL 都不一定逐点收敛**, 但一定均方收敛. (类似泰勒展开)

· Futhermore, Let Fourier Series of f is F. $F(x) \to f(x), f(x)$ continuous; $F(x) \to \frac{f(x^+)+f(x^-)}{2}, f(x)$ discontinuous.

Term-By-Term Integration: Fourier Series for *piecewise continuous* functions can be $\overline{\text{Integrated term-by-term}}$ to $\overline{\text{Convergent}}$ Fourier Series.

Finite Truncation (Partial Sum): Finite Truncation of Fourier Series is: $s_k = \frac{a_0}{2} + \sum_{n=1}^{k}\left(a_n cos(\frac{n\pi x}{L}) + b_n sin(\frac{n\pi x}{L})\right)$ || **Error**: $e_N = s_N - f(x)$

· $n \to \infty$: [1] Continuous points, Fourier approximation of $f(x)$ is better ($|e_n|$ decrease). [2] Discontinuous points, oscillation of Fourier series get smaller.

· **Gibbs Phenomenon**: The Amplitude of the *largest* oscillation (near discontinuous points) does *not decrease*.

4.3 Other Results on Fourier Series

Extension of a function: $f(x)$ know for $x \in [0, L]$ $\Rightarrow$ Even Ext.: $h(x) = \{f(x), [0, L]; \ f(-x), (-L, 0)$ || Odd Ext.: $h(x) = \{f(x), (0, L); \ 0, x = \pm L, 0; \ -f(-x), (-L, 0)$

Even Ext: $a_n = \frac{2}{L}\int_0^L f(x)cos(\frac{n\pi x}{L})dx, b_n = 0$ || Odd Ext: $b_n = \frac{2}{L}\int_0^L f(x)sin(\frac{n\pi x}{L})dx, a_n = 0$

Complex Form of Fourier Series: $f(x) = \sum_{-\infty}^{\infty} c_n e^{\frac{i\,n\pi x}{L}}$ || $c_n = \frac{a_n - ib_n}{2}; c_{-n} = \frac{a_n + ib_n}{2}; c_0 = \frac{a_0}{2}, n > 0$ || $a_0 = 2c_0; a_n = c_n + c_{-n}; b_n = i(c_n - c_{-n})$

· **Euler-Fourier Formula (Complex)**: If (⇑) of Periodic $f, (T = 2L)$ converges in $[-L, L]$ $\Rightarrow$ $c_n = \frac{1}{2L}\int_{-L}^{L} f(x)e^{-\frac{i\,n\pi x}{L}}$ || $c_{-n} = \overline{c_n}$ if f real.

Parseval's Theorem: If [1] Periodic function f; [2] f has convergent Fourier Series. $\Rightarrow$ Square Norm $(f, f) = \int_{-L}^{L} |f(x)|^2 dx = 2L\sum_{-\infty}^{\infty} |c_n|^2 = L\left[\frac{|a_0|^2}{2} + \sum_{n=1}^{\infty}(|a_n|^2 + |b_n|^2)\right]$ ps: 无限维的勾股定理

5 Linear PDEs by Separation of Variables

Requirement: Assume $u(x,t) = X(x)T(t)$! $\Rightarrow$ If PDE 能写成类似 $\frac{T'(t)}{T(t)} = \frac{X''(x)}{X(x)} \Rightarrow \frac{T'(t)}{T(t)} = \frac{X''(x)}{X(x)} = \lambda$ (Separation Constant) $\Rightarrow$ 可以使用此方法!

1.Homo, 假设: $u(0,t) = u(L,t) = 0; u(x, 0) = f(x)$ 2.Inhomo, 假设: change to $u(0,t) = a, u(L,t) = b$

Basic Idea|Homo: 1. 假设解 u 能写成 $u(x,t) = X(x)T(t)$ $\Rightarrow$ 2. 代入 PDE, 得两个 ODE $\Rightarrow$ 3. 解其中一个得到特征函数 X_n $\Rightarrow$ 4. 代 initial 条件得限制条件 $\Rightarrow$ 5. 解另一个得到特征函数 T_n $\Rightarrow$ 6. 特征函数 X_nT_n linear combination 得到 u. 7. 代 boundary 条件用 Euler-Fourier 得到的系数.

· Remark: [1] 化简中可以使用正比简化计算 [2] 对 n 的取值为 $1, 2, 3, \ldots$ [3] 对 λ 的取值让出现 sin, cos 时为正不容易出错, 或注意正负号

Basic Idea|Inhomo: Soln: $u(x,t) = v(x) + \omega(x,t), v(x) = \frac{b-a}{L}x + a$ || $\omega(x,t)$ 当作 homo 微分方程解, 边界条件 $\omega(x,0) = f(x) - v(x); \omega(0,t) = \omega(L,t) = 0$

Laplace Equation: 2D:$\nabla^2 u = \partial_x^2 u + \partial_y^2 u = 0$ 3D: $\nabla^2 u = \partial_x^2 u + \partial_y^2 u + \partial_z^2 u = 0$ Following are the method to solve:

1. **Rectangle**: $(0 \le x \le a, 0 \le y \le b)$ with $u(x, 0) = u(x, b) = u(0, y) = 0$; $u(a, y) = f(y)$ 直接按照 Homo 正常计算 ⇓ 可能用到 Euler Equation
2. **Circle**: $(u(r = a, \theta) = f(\theta)$; $u(r,\theta)$ $bounded)$ [1] $\partial_x^2 u + \partial_y^2 u = 0; x = rcos(\theta), y = rsin(\theta)$ 变极坐标 $R(r), \Theta(\theta)$ [2] 一样的计算 + 条件 $\Theta(\theta + 2\pi) = \Theta(\theta)$

Heat eqn: $\frac{\partial}{\partial t}u = \alpha^2\frac{\partial^2 u}{\partial x^2}$ - Parabolic

Wave eqn: $\frac{\partial^2}{\partial t^2}u = \alpha^2\frac{\partial^2 u}{\partial x^2}$ - Hyperbolic

Laplace eqn: $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ - Elliptical

一般来说都会有 bounded 的条件加以限制, 即如果存在 $r \to 0$ or $r \to \infty$ 会 unbounded 那么需要让其系数等于 0

6 Sturm-Liouville Theory (Another 'Fourier')

6.1 Sturm-Liouville Theory $Py'' + Qy' + Ry = 0 \Leftrightarrow [\mu Py']' + \mu Ry = 0$ Integrating factor:$\mu = \frac{1}{P}e^{\int_{x_0}^{x}\frac{Q(s)}{P(s)}ds}$

Sturm-Liouville (SL) Problem: Form: $-(p(x)y')' + q(x)y = \lambda r(x)y$, $p, r > 0$ || With: $a_1y(a) + a_2y'(a) = b_1y(b) + b_2y'(b) = 0$ p, p', q, r continuous. || **weight function**: $r(x)$. || ⇓**Lagrange's Identity**

· ⇒ Let *Linear Operator*: $\mathcal{L}(y) := -(p(x)y')' + q(x)y$ $\Rightarrow$ Problem becomes: $\mathcal{L}(y) = \lambda r(x)y$ (It's a Eigenvalue Problem $\mathcal{L}(\Psi_n) = \lambda_n r(x)\Psi_n$) $\Rightarrow$ $\forall$ solutions u, v with (二阶导连续) of SL Problem: $(\mathcal{L}(u), v) = (u, \mathcal{L}(v))$!!!

· **Properties|Real|Orthogonal**: In SL Problem: [1] λ_n, Ψ_n are *real*; || [2] Eigenfunctions Ψ_1, Ψ_2 with $\lambda_1 \ne \lambda_2$. **Modified Inner Product**: $\langle\Psi_1, \Psi_2\rangle = \int_a^b r(x)\Psi_1\Psi_2 dx = 0$ *i.e.*Ψ_1, Ψ_2 orthogonal.

· **Properties|Simple|Ordered**: In SL Problem: [3] λ_n are *simple*: $\forall\lambda_n$ has *unique* Ψ_n; || [4] λ_n are *Ordered*: $\lambda_1 < \lambda_2 < \cdots < \lambda_n < \cdots \to \infty$ as $n \to \infty$

Normalisation of Eigenfunction: [1] Ψ_n normalised if: $\langle\Psi_n, \Psi_n\rangle = 1$ || [2] Method (Normalised Ψ_n): $\Phi_n = \frac{1}{\sqrt{\langle\Psi_n, \Psi_n\rangle}}\Psi_n = [\int_a^b r(x)\Psi_n^2 dx]^{-1/2}\Psi_n$

Sturm-Liouville Convergence Theorem: If f, f' are piecewise continuous on $[0, 1]$ $\Rightarrow$ $F(x) = \sum_{n=1}^{\infty} c_n\Phi_n$ converges. || Φ_n are *orthonormal* eigenfunction of a Sturm-Liouville problem.

· Futhermore, [1] $F \to f$ when f is continuous; [2] $F \to \frac{f(x^+)+f(x^-)}{2}$ when f is discontinuous. 边界点右边界条件决定 **Calculate Coefficients**: $c_n = \langle f, \Phi_n\rangle = \int_a^b r(x)f(x)\Phi_n(x)dx$ || ps: c_n minimize the MSE ε_N

· 如果我们不需要归一化: 特征函数 Ψ_n $\Rightarrow$ $f(x) = \sum_{n=1}^{\infty} c_n\Psi_n$, $c_n = \langle f, \Psi_n\rangle = \int_a^b f(x)\Psi_n(x)dx$ || **Parseval's Identity**: f, f' are piecewise continuous on $[a, b]$ $\Rightarrow$ $\langle f, f\rangle = \int_a^b f^2(x)r(x)dx = \sum_{n=1}^{\infty} c_n^2$

6.2 General Boundary eigenvalue problem || Non-Homogeneous Boundary Value Problems || Open Interval Situation

General Boundary eigenvalue problem: $\mathcal{L}(y) = \lambda r(x)y$, $a < x < b$ || With $n-th$ order operator: $\mathcal{L}(y) = p_n(x)\frac{d^n y}{dx^n} + \cdots + p_1(x)\frac{dy}{dx} + p_0(x)y$ || Assume n linear Homogeneous Boundary conditions at the endpoints.

· ⇒ And suppose that $(\mathcal{L}(u), v) = (u, \mathcal{L}(v))$ ←(**Self-Adjoint Operator**)→ $\mathcal{L}$ must be <u>even order</u>. || **Properties**: [1] Infinite eigenvalues. [2] Orthogonal eigenfunctions. [3] 类似 Fourier 可以将任意函数用级数展开.

Non-Homogeneous BVP|Sturm-Liouville: $\mathcal{L}(y) = \mu r(x)y + f(x)$ where [1] $\mathcal{L}$ is Sturm-Liouville operator. (就是符合前面 SL Problem operator 的定义 + conditions) [2] μ is constant.

· ⇒ Let λ_n, Φ_n be eigenvalues & orthonormal eigenfunctions of $\mathcal{L}(y) = \lambda r(x)y$. || Method: **Green's Function**: $G(x,z) = \sum_{n=1}^{\infty}\frac{\Phi_n(x)\Phi_n(z)}{\lambda_n - \mu}$ $(\mu \ne \lambda_n)$ $\Rightarrow$ Solution: $y(x) = \int_a^b G(x,z)f(z)dz$ $(\mu \ne \lambda_n)$

· (Ctd.) [1] If $\mu \ne \lambda_n \Rightarrow$ unique solution. [2] If $\exists m, \mu = \lambda_m$: Let $c_n = \int_a^b f(x)\Phi_n(x)dx$ $\Rightarrow$ Case I ($c_m = 0$): Let $b_n = \frac{c_n}{\lambda_n - \mu}, n \ne m$ $y(x) = \sum_{n=1, n\ne m}^{\infty}[b_n\Phi_n(x)] + b_m\Phi_m(x), b_m \in \mathbb{R}$ Case II ($c_m \ne 0$): No Solution.

Sigular / Regular SL Problem: **Regular**: boundary condition at *closed interval*. || **Singular**: boundary condition at *open interval* or [a,b) or (a,b].

Sigular SL Problem|One-side: $\mathcal{L}(y) = \lambda r(x)y$, 奇异边 $x = a, b$. $\Rightarrow$ $\lim_{\varepsilon\to a,b} p(\varepsilon)[u'(\varepsilon)v(\varepsilon) - u(\varepsilon)v'(\varepsilon)] = 0$ $\Rightarrow$ $\star\,\forall$ sols u, v: $\int_a^b \mathcal{L}(u)v - u\mathcal{L}(v) = 0\,\star$ $\Leftarrow lim_{\varepsilon\to a,b}p(\varepsilon) = 0$ and u, u', v, v' bounded as $x \to a, b$

· ⋆ holds ⇒ [1] 特征值实数 [2] 正交特征函数 [3] 可以展开任意函数 || 解题一般思路: [1] 分离变量法得到 ODE [2] 解 ODE(观察是否是 Bessel 问题), 系数先不管 [3] 使用边界条件 (注意 bounded) 去除部分无关系数 [4] 通解: 特征函数线性组合 +series 方法确定系数

6.3 Other Results on Sturm-Liouville Theory

Mean Square Error: $\varepsilon_N = \int_a^b r(x)[f(x) - S_N(x)]^2$, (Truncation) $S_N(x) = \sum_{n=1}^{N} c_n\Phi_n(x)$ || If $\lim_{N\to\infty}\varepsilon_N = 0 \Rightarrow F = \sum_{n=1}^{\infty} c_n\Phi_n$ **converge in the mean** to f. (Not Pointwise Convergence)

Complete of Eigenfunctions: $\{\Phi_n\}$ (or $\{\Psi_n\}$) is complete with respect to [Mean Square Convergence] / [Pointwise Convergence] for a *class* of functions $\mathcal{F}$ if $[\lim_{N\to\infty}\varepsilon_N = 0]$ / $[\lim_{N\to\infty} S_N(x) = f(x)]$ for all $f \in \mathcal{F}$

Class of Square Integrable functions: $\mathcal{F} = \{f(x) | f \ and \ f^2 \ are \ integrable \ on \ x \in [a, b]\}$

Theorem|Completeness: The Set orthonormal eigenfunctions $\{\Phi_n\}$ of a *regular* SL boundary problem are *complete* with respect to [Mean Square Convergence]. for a *class* of Square Integrable functions.

7 Laplace's Transform 这里的 $\mathcal{L}$ 和前面无关

Laplace Transform: The Laplace transform of f is: $F(s) = \mathcal{L}\{f(t)\}(s) = \int_0^{\infty} e^{-st}f(t)dt$ || ps:May not exists

$f \in E$ **xponential Order**:= [1] f is piecewise continuous on $[0, \infty)$; [2] $\forall t \ge 0 : \exists A, B > 0$ s.t. $|f(t)| \le Ae^{Bt}$

Theorem|Laplace Existence: If $f \in E$ $\Rightarrow$ $\forall s$ sufficiently large, $\mathcal{L}\{f\}(s)$ exists.

Theorem|Linear Operator: $\mathcal{L}\{c_1f + c_2g\}(s) = c_1\mathcal{L}\{f\}(s) + c_2\mathcal{L}\{g\}(s)$ $\mathcal{L}\{0\} = 0$

Theorem|Derivative: If f continuous on $[0, \infty)$; $f, f' \in E$ $\Rightarrow$ $\mathcal{L}\{f'\}(s) = s\mathcal{L}\{f\}(s) - f(0)$

· $f, f', \ldots, f^{(n-1)} \in E$ and continuous $\Rightarrow$ $\mathcal{L}\{f^{(n)}\} = s^n\mathcal{L}\{f\}(s) - s^{n-1}f(0) - \cdots - sf^{(n-2)}(0) - s^0f^{(n-1)}(0)$

Inverse Laplace Transform: $f(t) = \mathcal{L}^{-1}\{F(s)\}(t) = \frac{1}{2\pi i}\int_{\gamma-i\infty}^{\gamma+i\infty} e^{st}F(s)ds$, $\Re(s) = \gamma$ (general form)

$f(t)$	$\mathcal{L}\{f\}(s)$	$f(t)$	$\mathcal{L}\{f\}(s)$	$f(t)$	$\mathcal{L}\{f\}(s)$
1	$\frac{1}{s}$	$t^n \mid e^{at}$	$\frac{n!}{s^{n+1}} \mid \frac{1}{s-a}, x > a$	t^ne^{at}	$\frac{n!}{(s-a)^{n+1}}$
$\ominus sinh(at)$	$\frac{a}{s^2-a^2}$	$sin(at)$	$\frac{a}{s^2+a^2}$	$cos(at)$	$\frac{s}{s^2+a^2}$
$\ominus cosh(at)$	$\frac{s}{s^2-a^2}$	$e^{at}sin(bt)$	$\frac{b}{(s-a)^2+b^2}$	$e^{at}cos(bt)$	$\frac{s-a}{(s-a)^2+b^2}$
$u_c(t)$	$\frac{e^{-sc}}{s}$	$t^ne^{at}sin(bt)$	$(-1)^n\frac{d^n}{ds^n}(\frac{b}{(s-a)^2+b^2})$	$t^ne^{at}cos(bt)$	$(-1)^n\frac{d^n}{ds^n}(\frac{s-a}{(s-a)^2+b^2})$

ps: 一般: $s > 0$ || $sinh, cosh$: $s > |a|$ || $e^{at}sin/cos(t^n)$: $s > a$ || ⋆ **Heaviside Unit Function**: $u_c = \chi_{[c,\infty)}$

Uniqueness & Existence: [1] If f, g continuous $\Rightarrow$ $\mathcal{L}\{f\} = \mathcal{L}\{g\} \Leftrightarrow f = g$

· [2] If f, g pointwise continuous ⇒ 唯一性不一定成立 (非连续点上不同) (No practical significance in applications)

Properties of Laplace Transform: Assume For $s > \gamma \ge 0$, $f \in E$ $\exists\,\mathcal{L}\{f(t)\}(s) = F(s)$

s-shift	$\mathcal{L}\{e^{ct}f(t)\}(s) = F(s-c), s - c > \gamma$	**s-derivatives**	$\mathcal{L}\{t^nf(t)\}(s) = (-1)^nF^{(n)}(s)$
t-shift	$\mathcal{L}\{f(t-c)u_c(t)\}(s) = e^{-sc}F(s)$	Corollary	$\mathcal{L}^{-1}\{e^{-sc}F(s)\} = f(t-c)u_c(t)$
scaling-1	$\mathcal{L}\{f(ct)\}(s) = \frac{1}{c}F(\frac{s}{c})$	**scaling-2**	$F(sc) = \frac{1}{c}\mathcal{L}\{f(\frac{t}{c})\}, c > 0$
δ | $\mathcal{L}\{\delta(t-t_0)\}(s) = e^{-st_0}, t_0 > 0; \lim_{t_0\to 0} e^{-st_0}, t_0 = 0$		δ / u_{t_0}	$[u_{t_0}(t)]' = \delta(t-t_0)$
Convolution	$\mathcal{L}\{(f\star g)(t)\}(s) = F(s)G(s) = \mathcal{L}\{f\}\mathcal{L}\{g\}$ $\mathcal{L}^{-1}\{F(s)G(s)\} = (f\star g)(t)$	$f\star g$	$(f\star g)(t) = \int_0^t f(t-\tau)g(\tau)d\tau$ $(f\star g)(t) = \int_0^t f(\tau)g(t-\tau)d\tau$

Impulse: $I(\tau) := \int_{t_0-\tau}^{t_0+\tau} g(t)dt$ (τ 很小) **Property**: [1] $I(\tau) = \int_{\mathbb{R}} g(t)dt = 1$ (or c) ; $\lim_{\tau\to 0} I(\tau) = 1$ (or c)

Dirac Delta Function: $\delta :=$ [1] $\delta(t - t_0) = 0, t \ne t_0$ [2] $\int_{\mathbb{R}}\delta(t - t_0)dt = 1$ it's a generalised function

· **Property**: [1] $\mathcal{L}\{\delta(t - t_0)\}(s) = \lim_{\tau\to 0}\mathcal{L}\{g(t - t_0)\} = e^{-st_0}$; [2] if f 连续, $\int_{\mathbb{R}}\delta(t - t_0)f(t)dt = f(t_0)$

Convolution Property: [1] $f\star(g + h) = f\star g + f\star h$ [2] $f\star g = g\star f$ [3] $f\star(g\star h) = (f\star g)\star h$ || NB: $f\star 1 \ne f$ and $f\star f \ne f^2$

Solving $ay'' + by' + cy = g(t)$ **with** $y(0) = \alpha, y'(0) = \beta$: **Transfer Function**: $H(s) = \frac{1}{as^2+bs+c}, \Phi(s) = \frac{(as+b)\alpha+a\beta}{as^2+bs+c}$

· soln: $y(t) = \phi(t) + \psi(t)$ || $\psi(t) = \int_0^t h(t-\tau)g(\tau)d\tau$, $\phi(t) = \mathcal{L}^{-1}\{\Phi(s)\}$ (ps:ϕ 是 homo 的解, 包括初始条件)

Related with integral: $\int_0^t f(\tau)d\tau = \int_0^t f(\tau)u_0(t-\tau)d\tau$ $\Rightarrow$ $\mathcal{L}\{\int_0^t f(\tau)d\tau\} = \frac{1}{s}\mathcal{L}\{f(t)\}$

8 Appendix

Wronskian Geometric Meaning: Geometrically, W is the volume spanned by the three vectors $y^{(1)}, y^{(2)}, y^{(3)}$; t 增加或减小代表在 $y^{(i)}$ 方向上 compress/stretch.

和差化积/积化和差: s+s=sc,s-s=cs,c+c=cc,c-c=-ss || e.g. $sinA + sinB = 2sin(\frac{A+B}{2})cos(\frac{A+B}{2})$ $sinAsinB = -\frac{1}{2}[cos(A+B) - cos(A-B)]$ || $sin(2x) = 2sin(x)cos(x)$ $cos(2x) = 2cos^2(x) - 1$ $cos(x) = \sqrt{\frac{1+cos(2x)}{2}}$ $sin(x) = \sqrt{\frac{1-cos(2x)}{2}}$

Hyperbolic Func: $sinh(x) = \frac{e^x - e^{-x}}{2}$ $cosh(x) = \frac{e^x + e^{-x}}{2}$ $cosh^2(x) - sinh^2(x) = 1$ $[sinh(x)]' = cosh(x)$ $[cosh(x)]' = sinh(x)$ || $\sinh(x \pm y) = \sinh(x)\cosh(y) \pm \cosh(x)\sinh(y)$ $\cosh(x \pm y) = \cosh(x)\cosh(y) \pm \sinh(x)\sinh(y)$

· $sinh(x)$ 单调递增, odd, iff $x = 0, sinh(x) = 0$ $cosh(x)$ $x > 0$ 单调递增, even || $sinh(2x) = 2sinh(x)cosh(x)$ $cosh(2x) = cosh^2(x) + sinh^2(x)$

Common Int: $\int tan(x)dx = -ln|cos(x)|$; $\int cot(x)dx = ln|sin(x)|$; $\int sec(x)dx = ln|sec(x) + tan(x)|$; $\int csc(x)dx = ln|csc(x) - cot(x)| = ln|tan(\frac{x}{2})|$ $[arcsin(x)]' = \frac{1}{\sqrt{1-x^2}}$; $[arccos(x)]' = -\frac{1}{\sqrt{1-x^2}}$

Common Der: $[tan(x)]' = sec^2(x)$; $[cot(x)]' = -csc^2(x)$; $[sec(x)]' = sec(x)tan(x)$; $[csc(x)]' = -csc(x)cot(x)$; $[arctan(x)]' = \frac{1}{1+x^2}$. || $1 + tan^2(x) = sec^2(x)$; $1 - csc^2(x) = cot^2(x)$; $csc^2(x) - cot^2(x) = 1$.

Polar Coordinate: $x = rcos(\theta), y = rsin(\theta)$ we have: $r\dot{r} = x\dot{x} + y\dot{y}$, $y\dot{x} - x\dot{y} = -r^2\dot{\theta}$

Explain of Fourier Series: In application, it's decomposition of signal into infinite (countable) waves; each wave $cos(\frac{n\pi x}{L}), cos(\frac{m\pi x}{L})$ has discrete period; a_n, b_m correspond to amplitudes of waves. || In vector algebra, it's a space of periodic functions; f corresponds to an arbitrary vector **v**; orthogonal basis $\{1, cos(\frac{n\pi x}{L}), sin(\frac{m\pi x}{L})\}$; a_n, b_m correspond to coefficient of each components of basis.

Even & Odd Function: Even: $f(-x) = f(x)$; Odd: $f(-x) = -f(x)$. || Property: [1] $f(x)$ odd: $\int_{-L}^{L} f(x)dx = 0$ [2] $f(x)$ even: $\int_{-L}^{L} f(x)dx = 2\int_0^L f(x)dx$ || [1] eve + eve = eve; [2] odd + odd = odd; [3] eve × eve = eve; [4] odd × odd = eve; [5] eve × odd = odd.

Euler Formula: $e^{ix} = cos(x) + isin(x)$ || $e^{inx} = cos(nx) + isin(nx)$ || $cos(nx) = \frac{e^{inx}+e^{-inx}}{2}$ || $sin(nx) = \frac{e^{inx}-e^{-inx}}{2i}$

Solve $y'' + \lambda y = 0$: [1] $\lambda < 0 : y = c_1\sinh(\sqrt{-\lambda}x) + c_2\cosh(\sqrt{-\lambda}x)$ [2] $\lambda = 0 : y = c_1x + c_2$ [3] $\lambda > 0(\mathbb{C}) : y = c_1\cos(\sqrt{\lambda}x) + c_2\sin(\sqrt{\lambda}x)$ || **Solve** $y' + \beta y = 0$: $y = c_1e^{-\beta x}$

Approximation $sin(x) \approx x$ when $x \to 0$ || ⇓一般对于 $-(xy')' = \lambda xy$ Let $t = \sqrt{\lambda}x \Rightarrow y'' + \frac{1}{t}y' + y = 0$

Bessel's Equation: $y'' + \frac{1}{t}y' + (1 - \frac{m^2}{t^2})y = 0$ $\Rightarrow$ Solution: $y(t) = c_1J_m(t) + c_2Y_m(t)$ || $J_m(t)$ is Bessel function of first kind; $Y_m(t)$ is Bessel function of second kind. || ps: $Y_m(t)$ not bounded as $t \to 0$, $J_m(t)$ bounded.

8.1 Some Applications

Generalised Heat Equation: $r(x)\partial_t u = \partial_x[p\partial_x u] - qu + F$ $\Rightarrow$ [1] 考虑 Homo: 分离变量, 得到一个 SL 问题; 一个简单 ODE.ODE 不计算. [2] 将 F/r 进行 SL 分解为 Series.[3] 考虑解 $u = \sum X_n(x)b_n(t), b_n$ 未知;2,3 步带入原始方程, 根据特征向量正交得系数关系: 一个 ODE.[4] 求解 ODE 系数由其他边界条件给出.